

SIMATS ENGINEERING ASSESSMENT TOOL 2

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA17

COURSE OUTCOME COVERED

CO1: Apply AI basics such as intelligent agents and uninformed search methods to solve problems effectively.(BL3)

Assessment Tool Weightage: 50%

QUESTIONS: 2

**Duration : 1 Hour
Total Marks:30**

Annexure A

Constraint-Based Problem Solving

Code of Conduct:

I SK.Arshad Basha (192312326) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

SK.Arshad Basha




4x4 mini-sudoku grid must be filled with numbers 1-4 such that constraints

1 - - 4

- 4 1 -

2 - 4 -

- 3 - 1

$R1C2 = 2$

$R1C3 = 3$

$R2C1 = 3$

$R2C4 = 2$

$R3C2 = 1$

$R3C4 = 3$

$R4C1 = 4$

$R4C3 = 2$

Blank cells: $R1C2, R1C3, R2C1, R2C4, R3C2, R3C4, R4C1, R4C3$

Apply Back tracking search

cell 1: $R1C2$

Row 1 = $\{1, -, - 4\}$ missing number = $\{2, 3\}$

Column 2 = $\{4, 3\}$ missing number = $\{1, 2\}$

~~Common value = 2~~

Assign

$R1C2 = 2$

cell 2: $R1C3$

Row 1 = $\{1, 2, -, 4\}$

missing number = 3

Assign

$$R1C3 = 3$$

cell 3 : R2C2

$$\text{column 1} = \{1, 2\}$$

$$\text{missing numbers} = \{3, 4\}$$

$$\text{Row 2} = \{4, 1\}$$

$$\text{missing number} = \{2, 3\}$$

$$\text{common} = 3$$

Assign

$$R2C2 = 3$$

1 2 3 4

3 4 1 -

2 - 4 -

- 3 - 1

cells : R3C2

$$\text{column 2} = \{2, 4, -3\}$$

$$\text{missing number} = 1$$

Assign

$$R3C2 = 1$$

$$\text{mn} 1 = \{1, 3, 2, -\}$$

$$\text{missing number} = 4$$

Assign

$$\text{RUCI} = 4$$

Grid

1 2 3 4

3 4 1 2

2 1 4 3

4 3 - 1

Final Sudoku

1 2 3 4

3 4 1 2

2 1 4 3

4 3 2 1

Assignment order

1. $\text{R1C2} = 2$

2. $\text{R1C3} = 3$

3. $\text{R2C1} = 3$

4. $\text{R2C4} = 2$

5. $\text{R3C2} = 1$

6. $\text{R3C4} = 3$

7. $\text{R4C1} = 4$

Back tracking:- No backtracking was required because every assignⁿ satisfied the constraints.

② Fire fighting Robot operates in a grid environment (6x6) It must move from start (s) to a fire outbreak (4).

sol: Grid

```
S . . X . .  
. R . X . .  
. R . . . .  
. X X R . .  
. . . . R .  
. . . . . G
```

Legend:-

S = Start

G = Goal

. = Normal cell (cost=1)

R = Risky cell (cost=3)

X = Blocked cell.

UCS Execution

Expand S

cost = 0

and $(1,0)$

Cost = 1

$(2,0) = 2$

Queue = $[(0,1)=1, (2,0)=2]$

Step-3

Expand $(0,1)$

Cost = 1

New node = $(0,2) = 2$

Queue = $[(2,0)=2, (0,2)=2]$

Step-4

Expand $(2,0)$

Cost = 2

possible moves:- $(2,1)$ Risky

Cost:-

$$2+3=5$$

Step-5

Expand goal

Goal reached.

Optimal path

S
↓
(1,0)
↓
(2,0)
→
(2,1) Risky
→
(3,2)
→
(2,3)
→
(2,4)
↓
(3,4)
↓
(4,4) Risky
→
(4,5)
↓
G

$$\text{Normal moves} = 8 \times 1 = 8$$

$$\text{Risky moves} = 2 \times 3 = 6$$

$$\text{Total path cost} = 8 + 6 = 14$$

∴ This is a complete, of the normal moves and the risky moves.